ELEC3104: Digital Signal Processing

Topic 4: Digital Resonator Fundamentals

Model Answers to Tutorial questions

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Topic 4: Tutorial Questions (SOLO -Level 1)

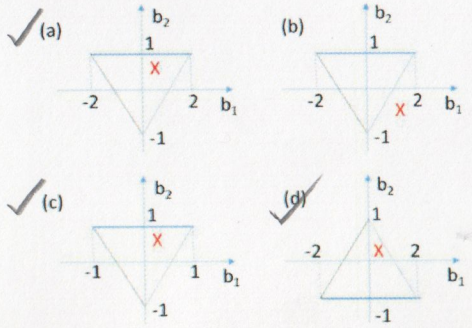
SOLO – Level 1

1. A second-order all-pole filter is described by $H(z) = \frac{1}{1+b_1 z^{-1}+b_2 z^{-2}}$

   Which of the following diagram(s) is used to conclude that the system is stable ? (Tick the correct answer)

The poles may be real or complex, but for a stable system, they must be inside the stability triangle.

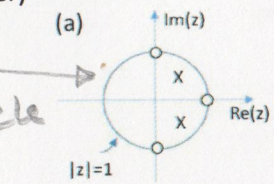
outside the stability triangle

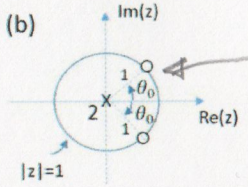


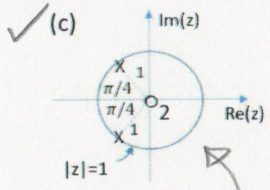
2. Which of the following pole-zero plot represents a digital oscillator? (Tick the correct answer)

poles inside the unit circle

no complex poles







(c) None of the above

Poles are on the unit circle

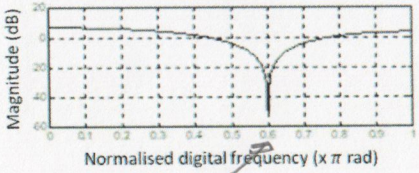
Topic 4: Tutorial Questions (SOLO -Level 1)

SOLO – Level 1

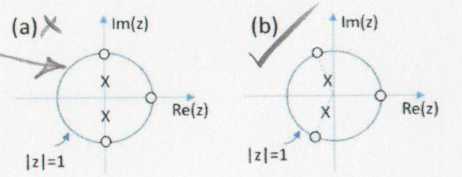
3. Which of the following pole-zero plot is most likely to produce the notch filter response shown below ?(Tick the correct answer)

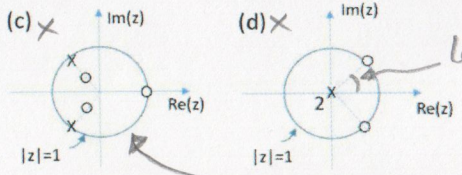
Notch frequency $\pi/2$

Notch filter magnitude response



notch frequency is > $\pi/2$





less than $\pi/2$

Zeros must be on the unit circle or closer to unit circle

4. An instrumentation and recording system require that the power-line frequency of 50Hz and harmonics to be eliminated. What type of filter would you choose? (tick the correct answer)

(a) All-pass filter
(b) Band-pass filter
✓(c) Notch filter ← To remove a particular frequency use notch filter.
(d) High-pass filter
(e) Low- pass filter
(f) An oscillator
(g) None of the above

Topic 4: Tutorial Questions (Level 2)

SOLO – Level 2

1. A digital oscillator has a unit impulse response given by:

$$h[n] = \sin([n+1]\theta_0)\,u[n];$$

where θ_0 is the oscillator frequency;

(a) Find the transfer function of this oscillator

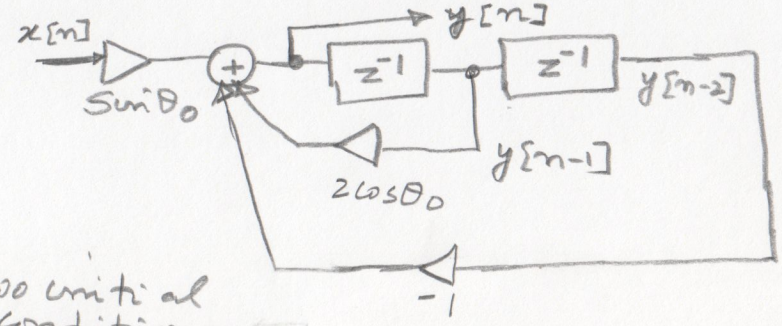
(b) Draw the structure for this transfer function using the transfer function obtained in part (a) above

(c) By setting the input in part (b) to zero and under certain initial conditions, sinusoidal oscillation can be obtained. Find these initial conditions.

(a)

From the z-transform table

$$H(z) = \frac{\sin\theta_0}{1 - 2\cos\theta_0 z^{-1} + z^{-2}}$$



$x[n] = 0$; $y[n] = \sin[(n+1)\theta_0]$

$n=0$ $\quad y[0] = \sin[(0+1)\theta_0] = \sin\theta_0$

$n=-1$ $\quad y[-1] = \sin[(-1+1)\theta_0] = 0$

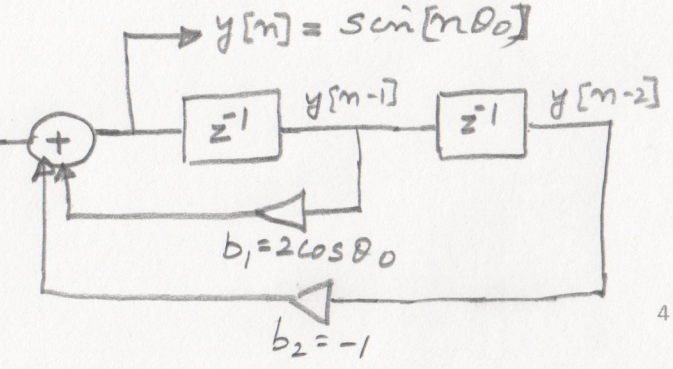
Two initial conditions

or

$n=-2$ $\quad y[-2] = \sin[(-2+1)\theta_0] = -\sin\theta_0$ ← Two initial conditions

2. Using the following difference equation, you are required to implement (i.e. draw a structure) a digital oscillator that generates y[n]=sin(θ_0n) for n ≥0. Choose appropriate initial conditions. Note that: $x[n]$ is the input; $y[n]$ is the output and r is a constant. You are required to choose an appropriate value for r in order to sustain oscillation.

$$y[n] = 2r\cos\theta_0\, y[n-1] - r^2 y[n-2] + x[n] - r\cos\theta_0\, x[n-1]$$



Let $x[n] = 0$, $\quad y[n] = 2r\cos\theta_0\, y[n-1] - r^2 y[n-2]$

For an oscillator $r = 1$ (Poles on the unit circle)

$\therefore y[n] = 2\cos\theta_0\, y[n-1] - y[n-2]$

output : $y[n] = \sin(\theta_0 n)$ ← given

$\therefore y[0] = \sin(0) = 0$ & $y[-1] = \sin(-\theta_0) = -\sin\theta_0$ ⇒

$y[0] = 0$
$y[-1] = -\sin\theta_0$

initial conditions

Topic 4: Tutorial Questions (Level 2)

SOLO – Level 2

3. Design a notch filter with notch frequency 60 Hz, stopband 5 Hz (bandwidth) and sampling frequency 300Hz.

$\Delta\theta_0 = 2\pi \frac{f_a}{f_s} = 2\pi \frac{5}{300} = \frac{\pi}{30}$ ($\Delta\theta_0$: bandwidth)

notch frequency $\theta_0 = 2\pi \times \frac{60}{300} = \frac{2\pi}{5}$

$r = 1 - 0.5\,\Delta\theta_0 = 1 - 0.5 \times \frac{\pi}{30} = 0.9476$

$\left(r = 1 - \pi \frac{\Delta f}{f_s}\right)$

→ Stopband: Place zeros at the notch frequency

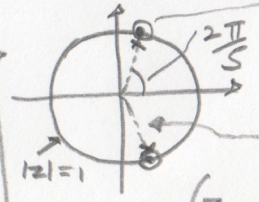
$z_1 = e^{+j\theta_0}$; $z_2 = e^{-j\theta_0}$ } zeros

$z_1 = e^{j\frac{2\pi}{5}}$; $z_2 = e^{-j\frac{2\pi}{5}}$

Passband: We place poles along the orientation of the zeros at

$P_{1,2} = r e^{\pm j\theta_0} \Rightarrow P_1 = 0.9476\, e^{j\frac{2\pi}{5}}$

$P_2 = 0.9476\, e^{-j\frac{2\pi}{5}}$



$\therefore H(z) = \frac{(z - e^{j\frac{2\pi}{5}})(z - e^{-j\frac{2\pi}{5}})}{(z - 0.9476\, e^{j\frac{2\pi}{5}})(z - 0.9476\, e^{-j\frac{2\pi}{5}})}$

$$H(z) = \frac{z^2 - 0.618z + 1}{z^2 - 0.5857z + 0.898}$$

4. Design a bandpass filter with centre frequency of 100Hz, passband 10Hz, stopband edges at 50 Hz and 150 Hz , and sampling frequency 400Hz.

Note: If the designed filter becomes non causal, then use double poles at each pole locations to get a causal filter.

$\theta_0 = 2\pi \frac{100}{400} = \frac{\pi}{2}$; $\Delta\theta_0 = 2\pi \frac{10}{400} = \frac{\pi}{20}$ ($\Delta\theta_0$: BW)

Stoband edges: $\theta_L = 2\pi \frac{50}{400} = \frac{\pi}{4}$; $\theta_u = 2\pi \frac{150}{400} = \frac{3\pi}{4}$

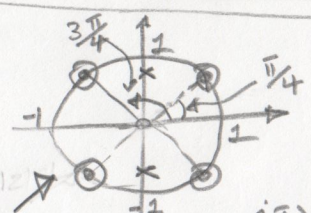
$r = 1 - 0.5\,\Delta\theta_0 = 1 - 0.5 \times \frac{\pi}{20} = 0.9125$

Passband: Place Poles at $P_{1,2} = r e^{\pm j\theta_0} = 0.9125\, e^{\pm j\frac{\pi}{2}}$

$= \pm j0.9125$

stopband: Place conjugate zeros at:

$z_{1,2} = e^{\pm j\frac{\pi}{4}}$ and $z_{3,4} = e^{\pm j\frac{3\pi}{4}}$



$H(z) = \frac{(z - e^{-j\frac{\pi}{4}})(z - e^{j\frac{\pi}{4}})(z - e^{-j\frac{3\pi}{4}})(z - e^{j\frac{3\pi}{4}})}{(z - 1\,0.9215)(z + j\,0.9215)}$

$$H(z) = \frac{z^4 + 1}{z^2 + 0.8941}$$

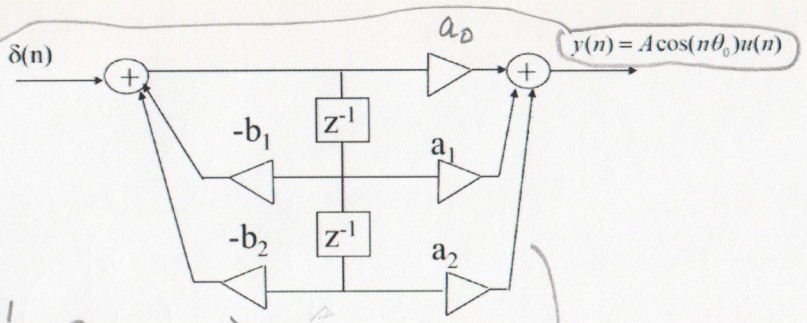
← This is a non-causal filter. To make it causal we could, for example, use double poles at each pole location to get a causal filter. $H(z) = \frac{z^4+1}{(z^2+0.8941)^2}$

Topic 4: Tutorial Questions (Level 2)

SOLO – Level 2

5. An IIR oscillator which is initialised by an impulse is shown below. The desired impulse response is :

$y[n] = A\cos[n\theta_0]u[n]$; where θ_0 is the oscillator frequency;



Assuming θ_0 is the oscillator frequency of the digital oscillator, and by writing appropriate equations, find the values of b_1, b_2, a_0, a_1 and a_2 for sustaining the oscillation.

$A\cos[n\theta_0]u[n] \xleftrightarrow{\text{Z-transform}} A\cdot\left(\frac{1-z^{-1}\cos\theta_0}{1-2\cos\theta_0 z^{-1}+z^{-2}}\right)$

$H(z) = \frac{a_0+a_1z^{-1}+a_2z^{-2}}{1+b_1z^{-1}+b_2z^{-2}}$

both equations are the same

By equating the coefficients we get

$a_0 = A$; $a_1 = -A\cos\theta_0$; $a_2 = 0$

$b_1 = -2\cos\theta_0$; $b_2 = 1$

6. The transfer function of a digital resonator with zeros at the origin is given by

$$H(z) = \frac{b_0}{(1-re^{j\theta_0}z^{-1})(1-re^{-j\theta_0}z^{-1})}$$

where r is the pole radius and θ_0 is the resonant frequency.

Show that the gain b_0 is given by the following, when $|H(\theta_0)| = 1$,

$$b_0 = (1-r)\sqrt{1+r^2-2r\cos(2\theta_0)}$$

$H(\theta) = \frac{b_0}{(1-re^{j\theta}e^{-j\theta})(1-re^{-j\theta_0}e^{-j\theta})}$

$|H(\theta)|_{\theta=\theta_0} = 1 = \left|\frac{b_0}{(1-re^{j\theta_0}e^{-j\theta_0})(1-re^{-j\theta_0}e^{-j\theta_0})}\right|$

$1 = \frac{b_0}{|(1-r)(1-re^{-j2\theta_0})|} = \frac{b_0}{(1-r)|1-r\cos 2\theta_0+jr\sin 2\theta_0|}$

$\therefore b_0 = (1-r)\sqrt{1+r^2-2r\cos(2\theta_0)} \leftarrow 1 = \frac{b_0}{1-r}\cdot\frac{1}{\sqrt{1+r^2-2r\cos 2\theta_0}}$

Topic 4: Tutorial questions (Level 3)

SOLO – Level 3

1. For the structure shown in Figure 1, write down the appropriate difference equations and hence state the function of this structure.

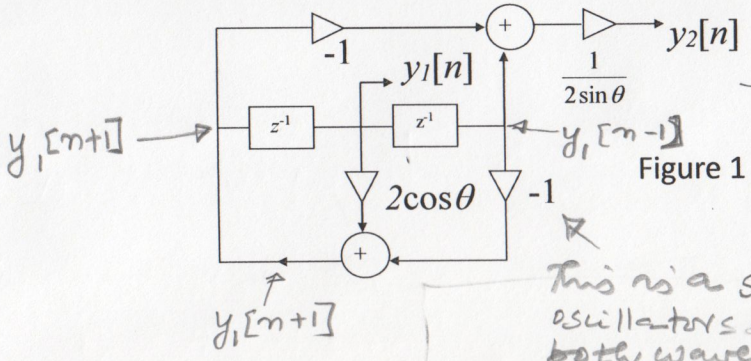


Figure 1

$y_2[n] = \frac{1}{2\sin\theta}\{y_1[n-1] - y_1[n+1]\}$ — ①

$y_1[n+1] = y_1[n]\cdot 2\cos\theta - y_1[n-1]$

$\therefore y_1[n] = \frac{1}{2\cos\theta}\{y_1[n+1] + y_1[n-1]\}$ — ②

If $y_1[n] = \cos(n\theta)$, then $y_2[n] = \sin(n\theta)$

Check: $\cos(n\theta) = \frac{1}{2\cos\theta}[\cos(n+1)\theta + \cos(n-1)\theta]$

$\therefore 2\cos\theta\cdot\cos n\theta = \cos(n+1)\theta + \cos(n-1)\theta$

This is correct

Similarly it can be shown, using equation (2)

$2\sin\theta\sin(n\theta) = \cos(n-1)\theta - \cos(n+1)\theta$

This is a sine & cosine oscillators delivering both waveforms simultaneously

2. The transfer function of a digital resonator with zeros at the origin is given by

$$H(z) = \frac{b_0(1-z^{-1})(1+z^{-1})}{(1-re^{j\theta_0}z^{-1})(1-re^{-j\theta_0}z^{-1})}$$

where r is the pole radius and θ_0 is the resonant frequency.
Determine the gain b_0 for the digital resonator so that $|H(\theta_0)| = 1$.

$$H(\theta) = \frac{b_0(1-e^{-j\theta})(1+e^{-j\theta})}{(1-re^{j\theta_0}e^{-j\theta})(1-re^{-j\theta_0}e^{-j\theta})}$$

$$|H(\theta)|_{\theta=\theta_0} = 1 = \frac{b_0|(1-e^{-j2\theta_0})|}{|(1-re^{j\theta_0}e^{-j\theta_0})(1-re^{-j\theta_0}e^{-j\theta_0})|}$$

$$1 = \frac{b_0|(1-\cos 2\theta_0) + (\sin 2\theta_0)|}{|(1-r)(1-re^{-j2\theta_0})|}$$

$$1 = \frac{b_0}{1-r}\cdot\frac{\sqrt{2-2\cos(2\theta_0)}}{\sqrt{1+r^2-2r\cos(2\theta_0)}} \qquad \sqrt{4\sin^2\theta_0} = 2|\sin\theta_0|$$

$$\therefore \boxed{b_0 = \frac{(1-r)\sqrt{1+r^2-2r\cos(2\theta_0)}}{2|\sin\theta_0|}}$$

Topic 4: Tutorial questions (Level 3)

SOLO – Level 3

3. Using the z-transform table, (a) determine the difference equations for the digital oscillators that have impulse responses $h[n] = A\sin[n\theta_0]u[n]$ and $h[n] = A\cos[n\theta_0]u[n]$ respectively; (b) Determine the structure for the coupled-form oscillator by combining the structure for the digital oscillator obtained in part (a) above.

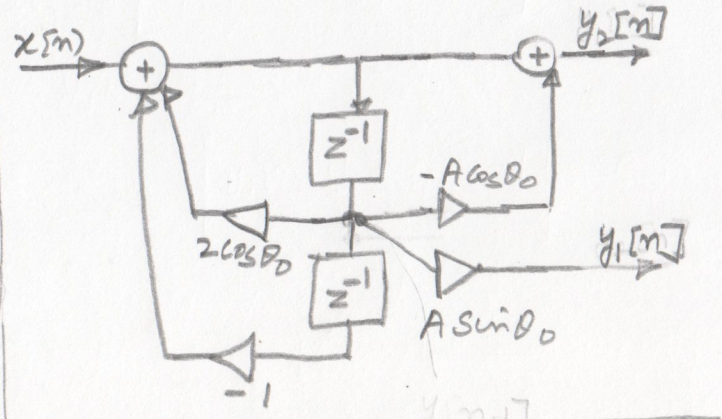
(a) $h[n] = A\sin(n\theta_0)u[n] \Rightarrow H(z) = A\dfrac{1 - z^{-1}\sin\theta_0}{1 - 2\cos\theta_0 z^{-1} + z^{-2}}$

Hence $y_1[n] = y[n] = 2\cos\theta_0\, y[n-1] - y[n-2] + A\sin\theta_0\, x[n-1]$ -①

$h[n] = A\cos(n\theta_0)u[n] \Rightarrow H(z) = A\dfrac{1 - z^{-1}\cos\theta_0}{1 - 2\cos\theta_0 z^{-1} + z^{-2}}$

Hence $y_2[n] = y[n] = 2\cos\theta_0\, y[n-1] - y[n-2] + A\,x[n] - A\cos\theta_0\, x[n-1]$ -②

(b) Combine equation ① & ② to obtain the structure.



4. Design an FIR filter that will reject a very strong 60-Hz sinusoidal interference contaminating a 200Hz useful sinusoidal signal. Determine the gain of the filter so that the useful signal does not change amplitude. The filter operates at a sampling frequency f_s = 500 samples/sec.

The rejected frequency $\theta_1 = 2\pi\frac{60}{500} = \frac{6\pi}{25}$

The frequency of useful signal $\theta_2 = 2\pi\frac{200}{500} = \frac{4\pi}{5}$

The useful signal does not change amplitude:

$\therefore |H(\frac{4\pi}{5})| = 1$ & $|H(\frac{6\pi}{25})| = 0$.

$\therefore H(\theta) = k_0\,(1 - 1e^{j\frac{6\pi}{25}}e^{-j\theta})(1 - 1e^{-j\frac{6\pi}{25}}e^{-j\theta})$

filter, gain, zero at the unit circle

$\rightarrow H(\theta) = k_0\left[1 - e^{-j\theta}\left(e^{j\frac{6\pi}{25}} + e^{-j\frac{6\pi}{25}}\right) + e^{-j2\theta}\right]$

$\therefore H(\theta) = k_0\, e^{-j\theta}\left[e^{j\theta} - 2\cos\frac{6\pi}{25} + e^{-j\theta}\right]$

$H(\theta) = k_0\, e^{-j\theta}\left(2\cos\theta - 2\cos\frac{6\pi}{25}\right)$

$|H(\theta)|_{\theta=\frac{4\pi}{5}} = 1 = k\,|e^{-j\theta}|\,|2\cos\frac{4\pi}{5} - 2\cos\frac{6\pi}{25}|$ ($|e^{-j\theta}| = 1$)

$\therefore k = \dfrac{\frac{1}{2}}{\cos\frac{6\pi}{25} - \cos\frac{4\pi}{5}}$

Topic 4: Tutorial questions (Level 3)

SOLO – Level 3

5. The transfer function of a digital resonator is given by

$$H(z) = \frac{b_0}{(1 - re^{j\theta_0}z^{-1})(1 - re^{-j\theta_0}z^{-1})}$$

The impulse response of this resonator can be found by partial fraction expansion . Show that the impulse response h[n] is given by:

$$h[n] = \frac{b_0 r^n}{\sin(\theta_0)} \sin[(n+1)\theta_0]\, u[n]$$

$$H(z) = \frac{b_0 z^2}{(z - re^{j\theta_0})(z - re^{-j\theta_0})}$$

We seek the expansion of the form

$$\frac{H(z)}{z} = b_0 \left\{ \frac{z}{(z - re^{j\theta_0})(z - re^{-j\theta_0})} \right\} = \frac{A}{z - re^{j\theta_0}} + \frac{A^*}{z - re^{-j\theta_0}}$$

where

$$A = \left. \frac{z}{z - re^{-j\theta_0}} \right|_{z = re^{j\theta_0}} = \frac{re^{j\theta_0}}{re^{j\theta_0} - re^{-j\theta_0}}$$

$$= \frac{re^{j\theta_0}}{2j\, r \sin\theta_0 \, e^{j\pi/2}}$$

$$A = \frac{e^{j(\theta_0 - \pi/2)}}{2\sin\theta_0}$$

$$\therefore A^* = \frac{e^{-j(\theta_0 - \pi/2)}}{2\sin\theta_0}$$

$$\therefore H(z) = \frac{b_0}{2\sin\theta_0} \left[\frac{z\, e^{j(\theta_0 - \pi/2)}}{z - re^{j\theta_0}} + \frac{z\, e^{-j(\theta_0 - \pi/2)}}{z - re^{-j\theta_0}} \right] = \frac{b_0}{2\sin\theta_0} \left[\frac{e^{j(\theta_0 - \pi/2)}}{1 - re^{j\theta_0}z^{-1}} + \frac{e^{-j(\theta_0 - \pi/2)}}{1 - re^{-j\theta_0}z^{-1}} \right]$$

$$h[n] = \frac{b_0}{2\sin\theta_0} \left[e^{j(\theta_0 - \pi/2)} \cdot r^n e^{j\theta_0 n} + e^{-j(\theta_0 - \pi/2)} r^n \cdot e^{-j\theta_0 n} \right]$$

$$= \frac{b_0}{\sin\theta_0} \cdot r^n \left[\frac{e^{j((n+1)\theta_0 - \pi/2)} + e^{-j((n+1)\theta_0 - \pi/2)}}{2} \right] = \frac{b_0 \cdot r^n}{\sin\theta_0} \cdot \underbrace{\cos((n+1)\theta_0 - \pi/2)}_{= \sin((n+1)\theta_0)}$$

$$h[n] = \frac{b_0 r^n}{\sin\theta_0} \sin((n+1)\theta_0)\, u[n]$$

Topic 4: Tutorial questions (Level 4)

SOLO – Level 4

1. For the system shown in figure 2,

   (a) Express $y_1[n]$ in terms of $y_1[n-1]$, $y_2[n-1]$, and $x[n]$; do the same for $y_2[n]$.

   (b) You may assume $A = \cos(\theta_0)$; $B = \sin(\theta_0)$; $y_1[-1] = \cos(-\theta_0)$; $y_2(-1) = \sin(-\theta_0)$. If $x[n] = 0$, show that : $y_1[n] = \cos(n\theta_0)$ and $y_2[n] = \sin(n\theta_0)$

   (c) Obtain the system transfer function

   $H_1(z) = \frac{Y_1(z)}{X(z)}$ and $H_2(z) = \frac{Y_2(z)}{X(z)}$ where A and B arbitrary values.

   (d) If $A = B = \frac{\sqrt{2}}{2}$, sketch the pole-zero pattern for $H_1(z)$ and $H_2(z)$

   (e) If $A = B = \frac{\sqrt{2}}{2}$ and $x[n] = \delta[n]$, find the impulse response $h_1[n]$.

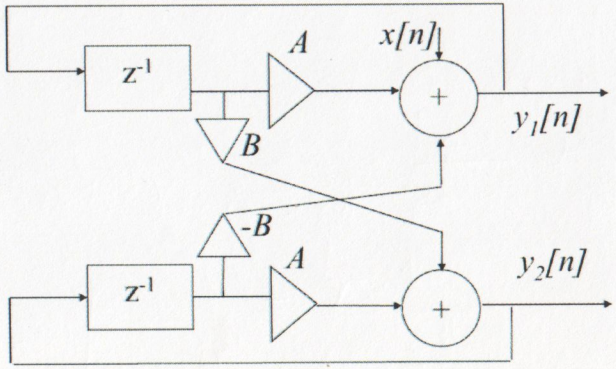


Figure 2

(a) $y_1[n] = x[n] + (-B)\, y_2[n-1] + A\, y_1[n-1]$

$y_2[n] = B\, y_1[n-1] + A\, y_2[n-1]$

(b) If $x[n] = 0$

$n=0$ $y_1[0] = \cos\theta_0\, y_1[-1] - \sin\theta_0\, y_2[-1]$
$= \cos\theta_0 \cos\theta_0 + \sin\theta_0 \sin\theta_0$
$= 1$

$n=1$ $y_1[1] = \cos\theta_0\, y_1[0] - \sin\theta_0\, y_2[0]$
$= \cos\theta$

$n=2$ $y_1[2] = \cos\theta_0\, y_1[1] - \sin\theta_0\, y_2[1]$
$= \cos\theta_0 \cos\theta_0 - \sin\theta_0 \sin\theta_0$
$= \cos^2\theta_0$

$n=n$ $\boxed{y_1[n] = \cos n\theta_0}$

$n=0$ $y_2[0] = \sin\theta_0\, y_1[-1] + \cos\theta_0\, y_2[-1]$
$= \sin\theta_0 \cos\theta_0 + \cos\theta_0 (-\sin\theta_0)$
$= 0$

$n=1$ $y_2[1] = \sin\theta_0\, y_1[0] + \cos\theta_0\, y_2[0]$
$= \sin\theta_0 \cdot 1$

$n=2$ $y_2[2] = \sin\theta_0\, y_1[1] + \cos\theta_0\, y_2[1]$
$= \sin\theta_0 \cos\theta_0 + \cos\theta_0 \cdot \sin\theta_0$
$= \sin 2\theta_0$

$n=n$ $\boxed{y_2[n] = \sin n\theta_0}$

(c) $Y_1(z) = X(z) - BY_2(z)z^{-1} + AY_1(z)z^{-1}$ ①

$Y_2(z) = BY_1(z)z^{-1} + AY_2(z)z^{-1}$

$\therefore Y_2(z)[1 - Az^{-1}] = BY_1(z)z^{-1}$ ②

Substitute in ①

$Y_1(z) = X(z) - Bz^{-1} \cdot \frac{BY_1(z)z^{-1}}{1 - Az^{-1}} + AY_1(z)z^{-1}$

$\therefore Y_1(z)\left[1 + \frac{B^2z^{-2}}{1 - Az^{-1}} - Az^{-1}\right] = X(z)$

$\therefore H_1(z) = \frac{1 - Az^{-1}}{1 - Az^{-1} + B^2z^{-2} - Az^{-1} + A^2z^{-2}}$

$\boxed{H_1(z) = \frac{1 - Az^{-1}}{1 - 2Az^{-1} + (A^2 + B^2)z^{-2}}}$

10 (a)

Topic 4: Tutorial questions (Level 4)

SOLO – Level 4

From ②, $Y_2(z) = \frac{B z^{-1}}{1 - A z^{-1}} \cdot Y_1(z) = \frac{B z^{-1}}{1 - A z^{-1}} \cdot \frac{(1 - A z^{-1}) \; X(z)}{1 - 2A z^{-1} + (A^2 + B^2) z^{-2}}$

$$\therefore \boxed{H_2(z) = \frac{Y_2(z)}{X_2(z)} = \frac{B z^{-1}}{1 - 2A z^{-1} + (B^2 + A^2) z^{-2}}}$$

(d) If $A = B = \frac{\sqrt{2}}{2}$, then $A + B = \sqrt{2}$ & $A^2 + B^2 = \frac{2}{4} + \frac{2}{4} = 1$

$$H_1(z) = \frac{1 - \frac{\sqrt{2}}{2} z^{-1}}{1 - 2 \times \frac{\sqrt{2}}{2} z^{-1} + 1 z^{-2}} = \frac{1 - \frac{\sqrt{2}}{2} z^{-1}}{1 - \sqrt{2} z^{-1} + z^{-2}}$$

$$\boxed{H_2(z) = \frac{\frac{\sqrt{2}}{2} z^{-1}}{1 - \sqrt{2} z^{-1} + z^{-2}}}$$

$$H_1(z) = \frac{z\left(z - \frac{\sqrt{2}}{2}\right)}{z^2 - \sqrt{2} z + 1} \rightarrow \text{Zeros: } z = 0; \; z = \frac{\sqrt{2}}{2}$$

$\rightarrow$ Poles: $P_{1,2} = \frac{1}{2}\sqrt{2}(1 \pm j)$

$$H_2(z) = \frac{\frac{\sqrt{2}}{2} z}{z^2 - \sqrt{2} z + 1}$$

Zeros: $z = 0$

Poles: $P_1 = \frac{1}{2}\sqrt{2}(1 + j)$; $P_2 = \frac{1}{2}\sqrt{2}(1 - j)$

(e) $H_1(z) = \frac{z^2 - z\frac{\sqrt{2}}{2}}{z^2 - \sqrt{2} z + 1}$

If $x[n] = \delta[n] \Rightarrow X(z) = 1 \Rightarrow Y_1(z) = \frac{z^2 - z\frac{\sqrt{2}}{2}}{z^2 - \sqrt{2} z + 1} \equiv \frac{z^2 - z\cos\theta_0}{z^2 - 2z\cos\theta_0 + 1}$ ← z-transform of $\cos(\theta_0 n)\, u[n]$

equate the coefficients

$\cos\theta_0 = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$

$\therefore \theta_0 = \pi/4$

$$\therefore y_1[n] = \cos\left[\frac{\pi}{4} n\right] u[n]$$ ← oscillator

10(b)

Topic 4: Tutorial questions (Level 4)

SOLO – Level 4

2. Consider a filter with transfer function

$$H(z) = \frac{b_0(1 - e^{j\theta_0}z^{-1})(1 - e^{-j\theta_0}z^{-1})}{(1 - re^{j\theta_0}z^{-1})(1 - re^{-j\theta_0}z^{-1})}$$

(a) Sketch the pole-zero pattern.

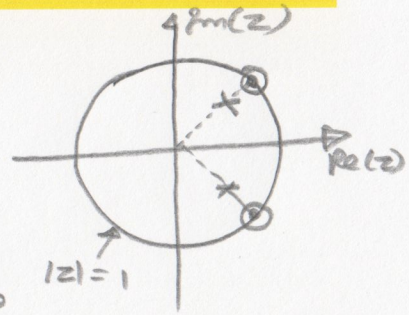
(b) Show that for $r \approx 1$, , the system is a notch filter and provide a rough sketch of its magnitude response if $\theta_0 = 60^o$.

(c) For $\theta_0 = 60^o$, choose b_0 so that the maximum value $|H(\theta)| = 1$.

(d) Determine the approximate 3-dB bandwidth of this filter.

(a) Poles : $p_{1,2} = re^{\pm j\theta_0}$

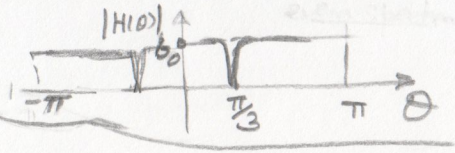
zeros : $z_{1,2} = e^{\pm j\theta_0}$



(b) for $\theta = \theta_0$, $H(\theta_0) = 0$

If $r \approx 1$, then $H(\theta) = b_0$ for $\theta \neq \theta_0$.

i.e Poles & zeros in H(z) cancel when $r \approx 1$



(c) $|H(\theta)|^2 = b_0^2 \dfrac{|1 - e^{j\theta_0}e^{-j\theta}|^2 \, |1 - e^{-j\theta_0}e^{j\theta}|^2}{|1 - re^{j\theta_0}\cdot e^{-j\theta}|^2 \, |1 - re^{-j\theta_0}\cdot e^{-j\theta}|^2}$

$= b_0^2 \dfrac{2(1 - \cos(\theta - \theta_0))}{1 + r^2 - 2r\cos(\theta - \theta_0)} \times \dfrac{2(1 - \cos(\theta + \theta_0))}{1 + r^2 - 2r\cos(\theta + \theta_0)}$ where $\theta_0 = \pi/3$ (given)

To find the Maximum value

$\dfrac{d|H(\theta)|^2}{d\theta} = 0 \Rightarrow \theta = \pi$ [you need to do the differentiation]

$\therefore |H(\pi)|^2 = 4b_0^2 \left(\dfrac{3}{1 + r + r^2}\right)^2 = 1 \Rightarrow$ $\boxed{b_0 = \frac{1}{3}(1 + r + r^2)}$

Topic 4: Tutorial questions (Level 4)

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(d) $|H(\theta)|^2 = b_0^2 \dfrac{|1-e^{-j\theta_0}.e^{-j\theta}|^2}{|1-re^{j\theta_0}.e^{-j\theta}|^2} \cdot \dfrac{|1-e^{-j\theta_0}.e^{j\theta}|^2}{|1-re^{-j\theta_0}.e^{-j\theta}|^2} = b_0^2 \dfrac{2[1-\cos(\theta-\theta_0)]}{1+r^2-2r\cos(\theta-\theta_0)} \times \dfrac{2[1-\cos(\theta+\theta_0)]}{[1+r^2-2r\cos(\theta+\theta_0)]}$

since the half-power frequencies are very close to θ_0, we have $\theta+\theta_0 \approx 2\theta_0$ & $r \approx 1$ and $\theta-\theta_0 \approx \frac{1}{2}\Delta\theta_0$ (↑ Bandwidth) ↑ assumption

$\therefore \quad \frac{1}{2} = b_0^2 \cdot \dfrac{2[1-\cos(0.5\Delta\theta_0)]}{1+r^2-2r\cos(0.5\Delta\theta_0)}$

$\therefore \cos(0.5\Delta\theta_0) = \dfrac{1+r^2-4b_0^2}{2r-4b_0^2}$

$\therefore \Delta\theta_0 = 2\cos^{-1}\left[\dfrac{1+r^2-4b_0^2}{2r-4b_0^2}\right]$ ↑ 3dB bandwidth

where $b_0^2 = \left[\frac{1}{3}(1+r+r^2)\right]^2$ (from part c)

Topic 4: Tutorial questions (Level 4)

SOLO – Level 4

3. By calculating the magnitude response of the following filter with a transfer function

$$H(z) = 1 + z^{-1} + z^{-2} + \ldots\ldots + z^{-8}$$

determine the frequencies of the analogue sinusoids that cannot pass through the filter. You may assume the sampling frequency f_s = 1 kHz.

$$H(z) = \frac{1 - z^{-9}}{1 - z^{-1}}$$

$$H(\theta) = \frac{1 - e^{-j9\theta}}{1 - e^{-j\theta}} = \frac{e^{-j9\theta/2}}{e^{-j\theta/2}} \cdot \frac{\left[e^{j9\theta/2} - e^{-j9\theta/2}\right]}{e^{j\theta/2} - e^{-j\theta/2}}$$

$$= e^{-j4\theta} \cdot \frac{\sin 9\theta/2}{\sin \theta/2}$$

$$|H(\theta)| = \left| \frac{\sin 9\theta/2}{\sin \theta/2} \right|$$

$H(\theta) = 0$ at $\theta = \frac{2\pi k}{9}$, $k = 1, 2, \ldots 8$

$$\theta = 2\pi \frac{f_a}{f_s} = \frac{2\pi k}{9}$$

$$\therefore f_a = \frac{k f_s}{9}$$

The corresponding analogue frequencies are, (1 kHz)

$$f_a = \frac{k f_s}{9} \quad k = 1, 2, 3, 4$$

or

$$f_a = \frac{1}{9}\text{ kHz}, \frac{2}{9}\text{ kHz}, \frac{3}{9}\text{ kHz}, \frac{4}{9}\text{ kHz}$$

Topic 4: Tutorial questions (Level 5)

SOLO – Level 5

1. Design a digital oscillator with adjustable phase, Ø , and produces an output

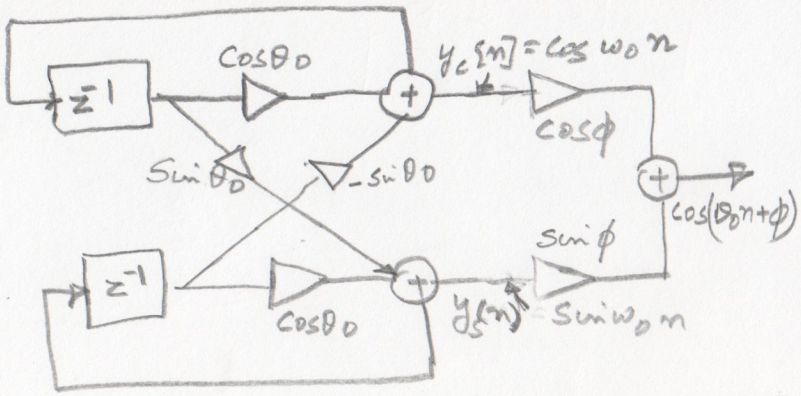
$$y[n] = \cos(n\theta_0 + \emptyset)\, u[n]$$

$h[n] = \cos(\omega_0 n + \phi)\, u[n]$

$= (\cos\omega_0 n \cos\phi - \sin\theta_0 n \cdot \sin\phi)\, u[n]$

Use the coupled-form oscillator (Page 17, Topic 4) and multiply the two outputs by $\cos\phi$ and $\sin\phi$ respectively and add the products

$y_c[n]\cdot\cos\phi + y_s[n]\sin\phi = \cos(\theta_0 n + \phi)$



2. A digital notch filter is required to remove an undesirable 60 Hz hum associated with a power supply in an ECG recording application. The sampling frequency used is fs = 500 samples/sec.

   (a) Design a second-order FIR notch filter and

   (b) A second-order pole-zero notch filter for this purpose

   In both cases choose the gain b_0 so that $|H(\theta)| = 1$ *for* $\theta = 0$.

(a) $\theta_0 = \frac{60}{500}\times 2\pi = \frac{6\pi}{50}$

$H(z) = \left(1 - e^{-j\frac{6\pi}{50}} z^{-1}\right)\left(1 - e^{-j\frac{6\pi}{50}} z^{-1}\right)\cdot b_0$ ← constant

second-order filter → $\boxed{H(z) = b_0\left(1 - 2\cos\frac{6\pi}{50} z^{-1} + z^{-2}\right)}$

$H(\theta) = b_0\left(1 - 2\cos\frac{6\pi}{50} e^{-j\theta} + e^{-j2\theta}\right)$

$H(\theta) = 2b_0 e^{-j\theta}\left(\cos\theta - \cos\frac{6\pi}{50}\right)$

$|H(\theta)|_{\theta=0} = 1 \quad \therefore 1 = 2b_0\left(1 - \cos\frac{3\pi}{25}\right)$

$\therefore \boxed{b_0 = \frac{1}{2\left(1 - \cos\frac{3\pi}{25}\right)}}$

(b) $H(z) = b_0 \dfrac{\left(1 - e^{j\frac{6\pi}{50}} z^{-1}\right)\left(1 - e^{-j\frac{6\pi}{50}} z^{-1}\right)}{\left(1 - re^{j\frac{6\pi}{50}} z^{-1}\right)\left(1 - re^{-j\frac{6\pi}{50}} z^{-1}\right)}$

$|H(\theta)|_{\theta=0} = 1 \quad = \dfrac{2b_0\left(1 - \cos\frac{6\pi}{50}\right)}{1 - 2r\cos\frac{6\pi}{50} + r^2}$

$\therefore \boxed{b_0 = \dfrac{1 - 2r\cos\frac{6\pi}{50} + r^2}{2\left(1 - \cos\frac{6\pi}{50}\right)}}$

Topic 4: Tutorial questions (Level 5)

SOLO – Level 5

3. A comb filter may be viewed as a notch filter in which the spectral nulls occur periodically across the frequency band. Comb filters find applications in a wide range of practical systems, for example, in the rejection of power-line harmonics. A comb filter can be described using a moving average FIR filter with the difference equation

$$y[n] = \frac{1}{M+1}\sum_{k=0}^{M} x[n-k]$$

(a) Show that the frequency response of the above filter is given by $H(\theta) = \frac{e^{-j\theta M/2}}{M+1}\frac{\sin\theta(\frac{M+1}{2})}{\sin(\frac{\theta}{2})}$

(b) Sketch the magnitude characteristic of the comb filter (for M= 10) that you obtained in part (a) above, but clearly illustrating the existence of the periodically spaced zeros at $\theta_k = \frac{2\pi k}{M+1}$ $for\ k = 1,2,\ldots.M.$

(a) $y[n] = \frac{1}{M+1}\left[x[n] + x[n-1] + x[n-2] + \cdots + x[n-M]\right]$

$H(z) = \frac{1}{M+1}\left[1 + z^{-1} + z^{-2} + \cdots + z^{-M}\right] = \frac{1}{M+1}\cdot\frac{1-z^{-(M+1)}}{1-z}$

$H(\theta) = \frac{1}{M+1}\cdot\frac{1-e^{-j(M+1)\theta}}{1-e^{-j\theta}} = \frac{1}{M+1}\cdot\frac{e^{-j(\frac{M+1}{2})\theta}}{e^{-j\frac{\theta}{2}}}\cdot\frac{e^{+j(\frac{M+1}{2})\theta} - e^{-j(\frac{M+1}{2})\theta}}{e^{j\frac{\theta}{2}} - e^{-j\frac{\theta}{2}}}$

$$\boxed{H(\theta) = \frac{e^{-j\frac{M\theta}{2}}}{M+1}\cdot\frac{\sin(\frac{M+1}{2}\theta)}{\sin\frac{\theta}{2}}} \Rightarrow H(\theta) = 0 \text{ at } \theta = \frac{2\pi k}{M+1} \text{ for } k = 1,2,\ldots M$$

(b) If $M = 10$ $\therefore \theta = \frac{2\pi}{11}\times k$

